US supply-chain networks, measured against strong regularity

Preview of the application section

1 Stand-ins for the main article

This section exists only so that the preview compiles. In the article, the labels `sec:srg`*,* `sec:johnson-hamming`*,* `sec:spectra` *and* `thm:walks` *point to the article's own sections on strongly regular graphs, Johnson and Hamming graphs, spectra, and the theorem that* $(A^k)_{ij}$ *counts walks of length* k*.*

Theorem 1.1. *For a graph with adjacency matrix* A*, the entry* $(A^k)_{ij}$ *is the number of walks of length* k *from* i *to* j*.*

2 An application: supply chains in the US economy

Every industry buys inputs from other industries, which buy inputs from others in turn. The resulting network of supply chains is recorded, for the whole US economy, in the input–output accounts of the Bureau of Economic Analysis, and it is the standard object for studying how a disruption in one industry travels to others [1, 3]. This section applies the article's machinery to that network in two ways. First, the theorem that powers of an adjacency matrix count walks (Theorem 1.1) has a weighted form, and in that form it *is* the standard description of how requirements travel along supply chains. Second, the definition of strong regularity becomes a measuring instrument: we build a graph from the network and measure how far it is from strongly regular, using random graphs and a strongly regular graph as benchmarks. Strongly regular graphs do not occur in economic data, and nothing here claims that they do.

Table 1 records how strong each connection is, so that a reader can tell a theorem from a modelling choice.

Table 1: Strength of each link between the article's mathematics and the supply-chain application.

Connection	Status	Support
Powers of A count weighted supply chains	Direct	Theorem 2.2 (proved here)
Total requirements $= \sum_k A^k$	Direct	Theorem 2.3 (proved here)
Average supply-chain length	Direct	Proposition 2.5 (proved here)
$(A^2_\theta)_{uv}$ counts common neighbours	Direct	Theorem 1.1
Distance from strong regularity, $\delta(G)$	Direct	Proposition 2.7 (proved here)
Erdős–Rényi graph as a null model	Benchmark	Proposition 2.10 (proved here)
Graphs with the same degree sequence	Benchmark	Computation
Strongly regular graph $H(2, 20)$	Benchmark	Corollary 2.1
Industries as stand-ins for firms	Approximation	Industries aggregate many firms

2.1 The data: requirements between industries

The Bureau of Economic Analysis records, for each year, how much of each commodity every industry buys (the *Use* table) and how much of each commodity every industry produces (the *Make* table), in millions of dollars [9]. We use the tables at producer prices, before redefinitions, in the 2017 industry classification, as redistributed in the U.S. Environmental Protection Agency's open-source package `useeior` [12, 15]. From them we form the standard industry-by-industry matrix [14, Ch. 5].

Definition 2.1 (Direct requirements)**.** Let U be the commodity-by-industry matrix of intermediate purchases, V the industry-by-commodity matrix of production, $g = V\mathbf{1}$ the vector of industry outputs and $q = V^{\mathsf{T}}\mathbf{1}$ the vector of commodity outputs. Every industry has $g_j > 0$; let $B = U\,\mathrm{diag}(g)^{-1}$, and let D have entries $d_{ic} = v_{ic}/q_c$, with $d_{ic} = 0$ when $q_c = 0$ (a few commodities, such as used goods, are produced by no industry). The *direct-requirements matrix* is $A = DB$. Its entry $a_{ij} \geq 0$ is the number of dollars of industry i's output used directly to produce one dollar of industry j's output.

Column j of A is industry j's recipe of intermediate inputs per dollar of output; the rest of each dollar pays for labour, capital, taxes and imports not produced by any US industry. We take the $N = 400$ industries with the largest gross output in 2017, out of 402, numbered $1, \ldots, N$ in order of their BEA industry code; this ordering is used for every matrix in this section. We work mainly with 2017, and repeat the analysis for 2012 at this level and for every year from 2012 to 2023 at the coarser level of 71 industries. In each Use table a handful of entries are negative (in the 2017 detail table, for example, entries for used and secondhand goods and for a few farm and food products); we set them to zero, so that A is entrywise non-negative. All computations in this section are on these public tables and are labelled as such. The code that produces every number, table and figure is publicly available.[1]

2.2 Supply chains are weighted walks

Regard A as a weighted adjacency matrix of a directed graph on the industries, with an arc $i \to j$ of weight a_{ij} whenever $a_{ij} > 0$: an arc points from a supplier to a buyer. A *supply chain of length k* from i to j is a walk $i = v_0 \to v_1 \to \cdots \to v_k = j$, and its *weight* is the product $a_{v_0 v_1} a_{v_1 v_2} \cdots a_{v_{k-1} v_k}$.

Theorem 2.2 (Weighted walks)**.** *For every $k \geq 1$, $(A^k)_{ij}$ is the sum of the weights of all walks of length k from i to j.*

Proof. For $k = 1$ the only walk of length one from i to j is the arc itself, of weight a_{ij}. Suppose the statement holds for k. Every walk of length $k+1$ from i to j is a walk of length k from i to some vertex ℓ, followed by the arc $\ell \to j$, and its weight is the product of the two weights. Summing over ℓ and using the induction hypothesis, $\sum_{\ell} (A^k)_{i\ell}\, a_{\ell j} = (A^{k+1})_{ij}$. □

With all weights equal to 1 this is Theorem 1.1. The weighted form has a direct reading: to make one dollar of j's output, industry j uses $a_{\ell j}$ dollars from each supplier ℓ, each of which uses $a_{i\ell}$ dollars of i's output per dollar, and so on back along the chain. So $(A^k)_{ij}$ is the amount of i's output, in dollars, that one dollar of j's output requires through chains with exactly $k-1$ intermediate industries.

[1] `https://github.com/desaidaksh1206-netizen/supply-chain-graphs-srg`

Theorem 2.3 (Total requirements)**.** *Let $A \geq 0$ have spectral radius $\rho(A) < 1$. Then $I - A$ is invertible,*

$$L = (I - A)^{-1} = I + A + A^2 + A^3 + \cdots, \tag{1}$$

and every entry of L is non-negative.

Proof. Because $\rho(A) < 1$, $A^k \to 0$ as $k \to \infty$ [8, Sec. 5.6]. For each k, $(I - A)(I + A + \cdots + A^k) = I - A^{k+1}$. If $(I - A)x = 0$ then $x = A^{k+1}x \to 0$, so $x = 0$ and $I - A$ is invertible. Multiplying by $(I - A)^{-1}$ gives $I + A + \cdots + A^k = (I - A)^{-1}(I - A^{k+1}) \to (I - A)^{-1}$. Every partial sum is non-negative, so the limit is. □

L is the Leontief inverse [11]. By Theorem 2.2, L_{ij} adds up the requirements of one dollar of j's output for i's output along supply chains of every length. Its column sums are the *output multipliers*: the total output, across the economy, needed to deliver one dollar of j's product to final users [14, Ch. 6].

The hypothesis $\rho(A) < 1$ is often checked through the following bound.

Proposition 2.4. *For any real square matrix, $\rho(A) \leq \max_j \sum_i |a_{ij}|$.*

Proof. Let $Av = \lambda v$ with $v \neq 0$, and write $\|x\|_1 = \sum_i |x_i|$. Then

$$|\lambda| \, \|v\|_1 = \|Av\|_1 \leq \sum_j |v_j| \sum_i |a_{ij}| \leq \Big(\max_j \sum_i |a_{ij}|\Big) \|v\|_1,$$

and dividing by $\|v\|_1 > 0$ gives the bound for every eigenvalue. □

Since column j of A sums to the share of j's output spent on intermediate inputs, one expects every column sum to be below 1. On our data that fails for exactly 1 industry, state and local government passenger transit: its value added is negative in the tables, and its intermediate purchases are 1.59 times the value of its output. Proposition 2.4 therefore does not apply. We compute $\rho(A)$ directly instead: $\rho(A) = 0.489$ for 2017 and 0.513 for 2012, so Theorem 2.3 holds. The partial sum $I + A + \cdots + A^{20}$ of (1) is within 1.5×10^{-7} of L, entry by entry.

Proposition 2.5 (Average supply-chain length)**.** *Let $A \geq 0$ with $\rho(A) < 1$ and $A \neq 0$, and let $s_k = \mathbf{1}^\mathsf{T} A^k \mathbf{1} / \mathbf{1}^\mathsf{T}(L - I)\mathbf{1}$ for $k \geq 1$. Then $\sum_{k \geq 1} s_k = 1$ and*

$$\sum_{k \geq 1} k \, s_k = \frac{\mathbf{1}^\mathsf{T}(L - I)L \, \mathbf{1}}{\mathbf{1}^\mathsf{T}(L - I)\mathbf{1}}.$$

Proof. By (1), $L - I = \sum_{k \geq 1} A^k$, which gives the first claim. By Gelfand's formula $\rho(A) = \lim_{k \to \infty} \|A^k\|^{1/k}$ [8, Sec. 5.6], for any r with $\rho(A) < r < 1$ there is a constant C with $\|A^k\| \leq C r^k$ for all k, so $\sum_{k \geq 1} kA^k$ converges absolutely. Its terms are non-negative, so it can be rearranged: $\sum_{k \geq 1} kA^k = \sum_{k \geq 1} \sum_{j \geq k} A^j = \sum_{k \geq 1} A^k L = (L - I)L$, since $\sum_{j \geq k} A^j = A^k L$. □

So s_k is the share of all requirements, direct and indirect, that travels along chains of exactly k steps, and $\sum_k k s_k$ is the average length of a supply chain weighted by the dollars it carries; it is an economy-wide version of the average propagation length of [4].

Figure 1 shows the result (computation). Direct purchases carry $s_1 = 48.3\%$ of all requirements, two-step chains 25.5%, three-step chains 13.1%, and chains of four or more steps the remaining 13.1%. The average supply-chain length is 2.04 steps (2.13 in 2012). Successive shares shrink by a nearly constant factor: s_{k+1}/s_k lies between 0.490 and 0.528 for $k = 1, \ldots, 8$, close to $\rho(A) = 0.489$.

This is what one expects: by Perron–Frobenius theory, $\rho(A)$ is itself an eigenvalue of the non-negative matrix A, with a non-negative eigenvector [8, Sec. 8.3], and the largest eigenvalue sets the rate at which A^k decays. Output multipliers range from 1.00 to 4.26, with mean 2.07: on average, delivering a dollar of final product requires about two dollars of output across the economy.

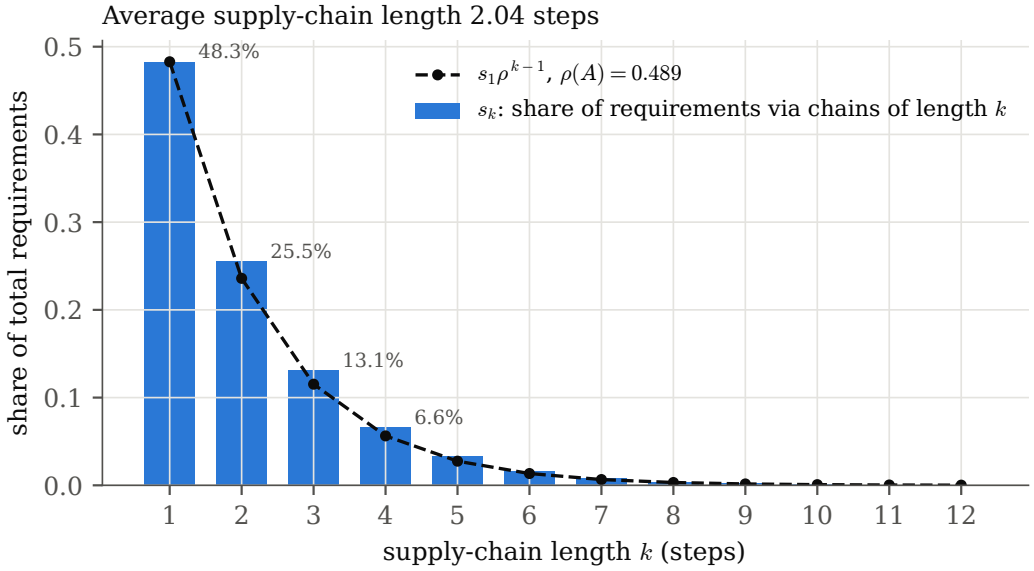


Figure 1: Share s_k of all requirements carried by supply chains of exactly k steps (bars), with the geometric decay $s_1\rho(A)^{k-1}$ (dashed), for the 400 industries in 2017. Computation on the BEA tables of Section 2.1.

2.3 From requirements to a graph

To ask how *structured* the network is, we turn A into a simple graph of the kind studied in this article.

Definition 2.6 (Threshold graph)**.** Let $w_{ij} = \max(a_{ij}, a_{ji})$ for $i \neq j$. For $\theta > 0$, the *threshold graph* G_θ has vertex set $\{1, \dots, N\}$, with $i \neq j$ adjacent if and only if $w_{ij} \geq \theta$, that is, if one of the two industries buys at least θ dollars from the other per dollar of its own output. Its adjacency matrix is A_θ.

Lowering θ adds edges and never removes them. No threshold is privileged, so the two measures defined in Section 2.4, δ and $\bar{\lambda} - \bar{\mu}$, are reported over 49 thresholds spaced evenly on a logarithmic scale from 10^{-4} (a hundredth of a cent per dollar) to 10^{-1} (ten cents per dollar). The other quantities are reported at the threshold at which G_θ has the density of $H(2, 20)$, defined in Section 2.5.

2.4 Measuring distance from strong regularity

Let G be a graph on n vertices with adjacency matrix A, and write $\bar{A} = J - I - A$ for the adjacency matrix of its complement. (In this subsection and the next, A is a 0/1 adjacency matrix.) By Theorem 1.1, for $u \neq v$ the entry $(A^2)_{uv}$ is the number of common neighbours of u and v; we write

$c(u,v) = (A^2)_{uv}$. The diagonal entry $(A^2)_{uu}$ is the degree $d(u)$. Recall from Section 1 that G, neither complete nor edgeless, is strongly regular with parameters (n,k,λ,μ) exactly when

$$A^2 = kI + \lambda A + \mu \bar{A}. \tag{2}$$

The three matrices I, A, $\bar{A}$ have disjoint supports: I covers the diagonal, A the ordered adjacent pairs, $\bar{A}$ the ordered non-adjacent pairs. Equation (2) therefore says three things at once: every diagonal entry of A^2 equals k, every entry on an edge equals λ, and every entry on a non-edge equals μ. For a general graph these entries are not constant. They form three lists:

$$\mathcal{D} = \big(d(u)\big)_u, \qquad \mathcal{L} = \big(c(u,v)\big)_{u\sim v}, \qquad \mathcal{M} = \big(c(u,v)\big)_{u\not\sim v,\, u\neq v}, \tag{3}$$

indexed by vertices, ordered adjacent pairs, and ordered distinct non-adjacent pairs respectively. If G has m edges and $\bar{m}$ non-edges, the lists have lengths n, $2m$ and $2\bar{m}$. Write $\bar{d}, \bar{\lambda}, \bar{\mu}$ for their means and Var for the population variance of a list.

Proposition 2.7. *Let G have at least one edge and at least one non-edge. For real k,λ,μ let $R(k,\lambda,\mu) = A^2 - kI - \lambda A - \mu\bar{A}$. Then*

$$\min_{k,\lambda,\mu\in\mathbb{R}} \|R(k,\lambda,\mu)\|_F^2 = n\,\mathrm{Var}(\mathcal{D}) + 2m\,\mathrm{Var}(\mathcal{L}) + 2\bar{m}\,\mathrm{Var}(\mathcal{M}), \tag{4}$$

and the minimum is attained exactly at $(k,\lambda,\mu) = (\bar{d},\bar{\lambda},\bar{\mu})$.

Proof. Because I, A and $\bar{A}$ have disjoint supports that together cover every entry, the squared Frobenius norm splits as

$$\|R\|_F^2 = \sum_u \big(d(u)-k\big)^2 + \sum_{u\sim v} \big(c(u,v)-\lambda\big)^2 + \sum_{u\not\sim v,\, u\neq v} \big(c(u,v)-\mu\big)^2,$$

with the last two sums over ordered pairs. Each term involves only one of k,λ,μ, so each can be minimised separately. For a list $x_1,\dots,x_\ell$ and real a, $\sum_i (x_i-a)^2 = \ell\,\mathrm{Var}(x) + \ell(\bar{x}-a)^2$, which is minimised exactly at $a=\bar{x}$ with minimum $\ell\,\mathrm{Var}(x)$. Applying this to each list gives (4). □

Definition 2.8 (Distance from strong regularity)**.** For G with at least one edge and one non-edge, set

$$\delta(G) = \frac{\min_{k,\lambda,\mu}\|R(k,\lambda,\mu)\|_F}{\|A^2\|_F}.$$

Taking $k=\lambda=\mu=0$ gives $R=A^2$, so $0 \le \delta(G) \le 1$. The normalisation makes δ comparable across thresholds, where the number of edges changes.

Corollary 2.1. *Let G be neither complete nor edgeless. Then $\delta(G) = 0$ if and only if G is strongly regular, in which case $(\bar{d},\bar{\lambda},\bar{\mu})$ are its parameters (k,λ,μ).*

Proof. By Proposition 2.7, $\delta(G) = 0$ if and only if all three lists in (3) are constant, which is (2) with $(k,\lambda,\mu) = (\bar{d},\bar{\lambda},\bar{\mu})$. □

The degree term in (4) cannot be dropped: constant λ and constant μ do not by themselves force a graph to be regular.

Example 2.9 (The bow-tie)**.** Let G be two triangles sharing one vertex, with vertices ordered v_0, v_1, v_2, v_3, v_4, where v_0 is the shared vertex and the triangles are $\{v_0, v_1, v_2\}$ and $\{v_0, v_3, v_4\}$. Then

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 4 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 2 \end{pmatrix}.$$

Every off-diagonal entry of A^2 is 1, so $\mathcal{L}$ and $\mathcal{M}$ are constant with $\lambda = \mu = 1$. The degrees are $4, 2, 2, 2, 2$, so $\bar{d} = 12/5$, $\mathrm{Var}(\mathcal{D}) = 32/5 - (12/5)^2 = 16/25$, and by (4) the minimum squared residual is $5 \cdot 16/25 = 16/5$. The graph is not strongly regular, and only the degree term detects it. By the friendship theorem of Erdős, Rényi and Sós [6], the finite graphs in which every two distinct vertices have exactly one common neighbour are precisely the graphs consisting of triangles sharing a common vertex, so the bow-tie is the smallest graph with $\lambda = \mu = 1$ that is not regular. Other values give smaller examples: the path $v_1v_2v_3$ has $\lambda = 0$ and $\mu = 1$, with degrees $1, 2, 1$.

2.5 The benchmarks

Random graphs. The random graph $G(n, p)$ of Gilbert [7], usually called the Erdős–Rényi model, includes each of the $\binom{n}{2}$ possible edges independently with probability p. Erdős and Rényi [5] studied the closely related model with a fixed number of edges.

Proposition 2.10. *In $G(n, p)$, each degree $d(u)$ has distribution $\mathrm{Bin}(n-1, p)$. For distinct u, v, the count $c(u, v)$ has distribution $\mathrm{Bin}(n-2, p^2)$ and is independent of whether u and v are adjacent.*

Proof. The degree of u counts which of the $n-1$ independent edges at u are present. For $u \neq v$, a third vertex w is a common neighbour exactly when both uw and vw are edges, which has probability p^2; these events for the $n-2$ choices of w involve disjoint pairs of edges and so are independent. None of them involves the edge uv, so $c(u, v)$ is independent of it. □

So in a random graph, adjacency tells us nothing about common neighbours: $\mathcal{L}$ and $\mathcal{M}$ have the same distribution, and $\bar{\lambda} - \bar{\mu}$ is close to zero.

Graphs with the same degrees. A random graph differs from G_θ in two ways at once: its degrees are nearly equal, and its edges are placed independently. To separate the two, we also compare with random graphs that have *exactly* the degree sequence of G_θ, generated by repeated double-edge swaps [13]: two edges $\{a, b\}$ and $\{c, d\}$ are replaced by $\{a, d\}$ and $\{c, b\}$ whenever this creates no loop or repeated edge, which leaves every degree unchanged. We make five swap attempts per edge. This benchmark is a computation; we prove nothing about it.

For each θ we report the mean and the central 95% range of δ and of $\bar{\lambda} - \bar{\mu}$ over 200 Erdős–Rényi graphs with the density of G_θ, and over 20 degree-preserving rewirings of G_θ (50 at θ^* below). These ranges are computations, not theorems.

A strongly regular graph. By Corollary 2.1, every strongly regular graph has $\delta = 0$. For the spectral comparison we want a specific strongly regular graph of the same order. Choosing $N = 400$ makes one available from Section 1: the Hamming graph $H(2, 20)$ [10, Example 2.2.2], whose vertices are the pairs (a, b) with $a, b \in \{0, \ldots, 19\}$, adjacent when they differ in exactly one coordinate. Its parameters are $(n, k, \lambda, \mu) = (400, 38, 18, 2)$, its non-trivial eigenvalues are the roots of $x^2 - (\lambda - \mu)x + (\mu - k) = x^2 - 16x - 36$ [2], namely 18 and -2, with multiplicities 38 and 361

(one checks $38 + 38 \cdot 18 - 361 \cdot 2 = 0 = \operatorname{tr} A$), and its largest cliques are its 40 "lines" of 20 vertices. We write θ^* for the threshold at which G_θ has exactly the density of $H(2, 20)$, hence mean degree 38; on our data $\theta^* = 0.0047$, that is, 0.47 cents per dollar.

2.6 Results

All numbers and figures in this subsection are computations on the 2017 BEA tables, except where another year is named. Figure 2 shows the adjacency matrix of G_{θ^*}, first in the ordering by industry code used throughout and then with the industries grouped by sector. The two matrices are A_{θ^*} and $PA_{\theta^*}P^{\mathsf{T}}$ for a permutation matrix P, so they describe the same graph.

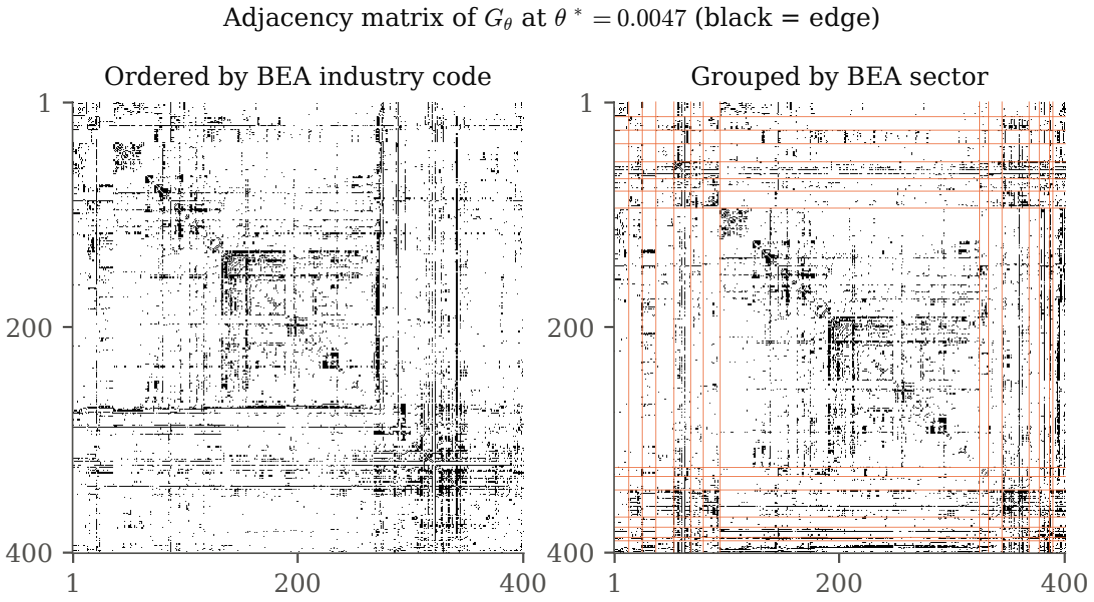


Figure 2: The adjacency matrix of G_{θ^*}, $\theta^* = 0.0047$, with industries in order of BEA code (left) and grouped by BEA sector (right; lines mark sector boundaries). Black entries are 1. The dense rows and columns are the hub industries, which trade with most other industries.

The graph at θ^*. G_{θ^*} has 7600 edges and mean degree 38.0, but its degrees are very unequal: their standard deviation is 37.6, against 5.3 for a random graph of the same density, and the largest is 323. The highest degrees belong to industries that trade with most other industries: Management of companies and enterprises (323), Truck transportation (289), Other real estate (279) and Electric power generation, transmission, and distribution (239). They appear as the dense rows and columns in Figure 2. The graph has 3 components; the 2 isolated vertices are BEA's "customs duties" and "private households" industries, which have no intermediate inputs. Its clique number is 14. Edges are only mildly concentrated inside the fifteen BEA sectors: 37% of edges lie inside a sector, against 34% of all pairs of industries.

Common neighbours. Figure 3 applies the instrument of Section 2.4 to G_{θ^*}. In the random graph, $\mathcal{L}$ and $\mathcal{M}$ have the same distribution, as Proposition 2.10 requires, with mean $(N-2)p^2 = 3.61$. In the industry network they separate: adjacent pairs have on average $\bar{\lambda} = 17.17$ common

neighbours and non-adjacent pairs $\bar{\mu} = 6.00$, a gap of $\bar{\lambda} - \bar{\mu} = 11.17$. The gap lies in $[-0.08, 0.07]$ for random graphs of this density and in $[9.32, 9.77]$ for graphs with the network's own degrees; for $H(2, 20)$ it is $\lambda - \mu = 16$. So adjacency predicts shared trading partners far better than in a random graph, and somewhat better than the unequal degrees alone would produce, but less sharply than in the strongly regular benchmark. Figure 4(b) shows that the order holds at every threshold: the gap exceeds both random ranges for every θ in $[0.0001, 0.100]$.

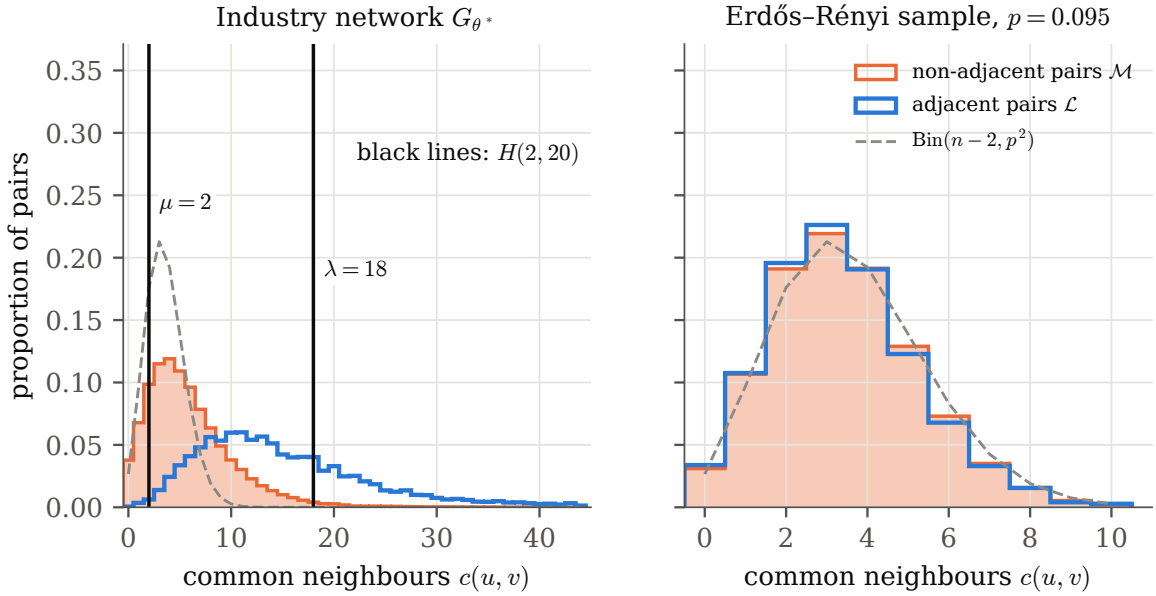


Figure 3: Distributions of the common-neighbour counts $c(u, v)$ for adjacent pairs ($\mathcal{L}$, blue) and non-adjacent pairs ($\mathcal{M}$, orange), in G_{θ^*} (left) and in one Erdős–Rényi graph of the same density (right), with the distribution Bin$(N-2, p^2)$ that Proposition 2.10 predicts for both lists in the random graph (dashed). Black lines: the constant values $\lambda = 18$ and $\mu = 2$ of $H(2, 20)$.

Distance from strong regularity. At θ^*, $\delta(G_{\theta^*}) = 0.649$. By Proposition 2.7 the squared residual splits into three parts: 47% from the spread of $\mathcal{L}$ (standard deviation 15.22), 45% from the spread of $\mathcal{M}$ (standard deviation 4.84), and only 8% from the unequal degrees, large as they are. Random graphs of the same density give δ in $[0.418, 0.433]$, and graphs with the same degrees give $[0.660, 0.665]$. The industry network is therefore further from strong regularity than a random graph of its density, but slightly closer than a random graph with its degrees. Figure 4(a) shows the same order across the sweep: $\delta(G_\theta)$ lies above the random-graph range for every θ in $[0.0001, 0.100]$ and below the degree-preserving range for every θ in $[0.0001, 0.087]$.

Spectra. Figure 5 compares the adjacency spectra at θ^*. $H(2, 20)$ has three distinct eigenvalues; the industry network has 399 (to within 10^{-6}) and the random graph 400, so on this count the industry network is as far from strong regularity as a random graph. Its largest eigenvalue is 62.2, against 38.4 for the random graph: a largest eigenvalue well above the mean degree 38 is what very unequal degrees produce, since it lies between the mean and the maximum degree (Section 1).

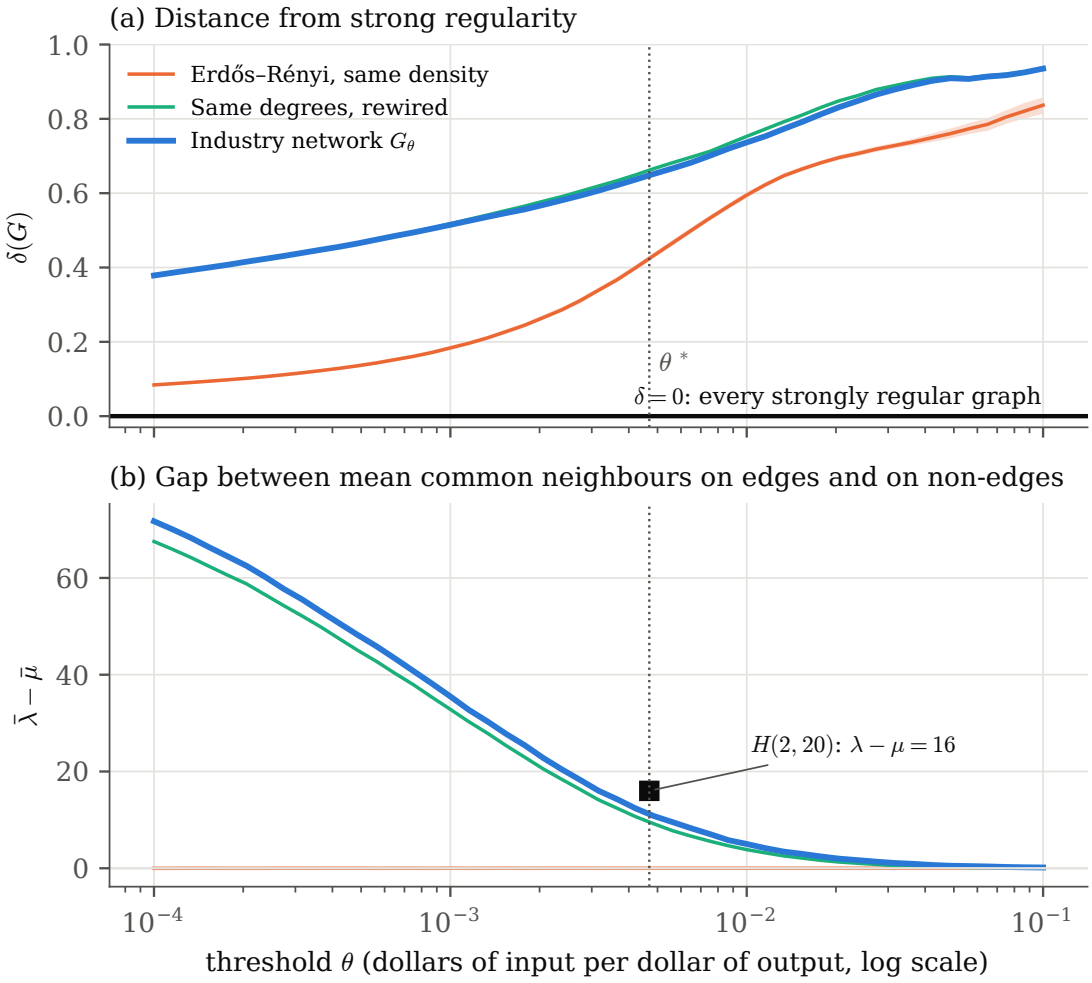


Figure 4: (a) Distance from strong regularity $\delta(G_\theta)$ as the threshold varies, for the industry network (blue), random graphs of the same density (orange) and random graphs with the same degree sequence (green); shading is the central 95% range of the benchmarks. The line $\delta = 0$ is every strongly regular graph. (b) The gap $\bar{\lambda} - \bar{\mu}$ for the same graphs, and $\lambda - \mu = 16$ for $H(2, 20)$. The dotted line marks θ^*. The horizontal axis is logarithmic.

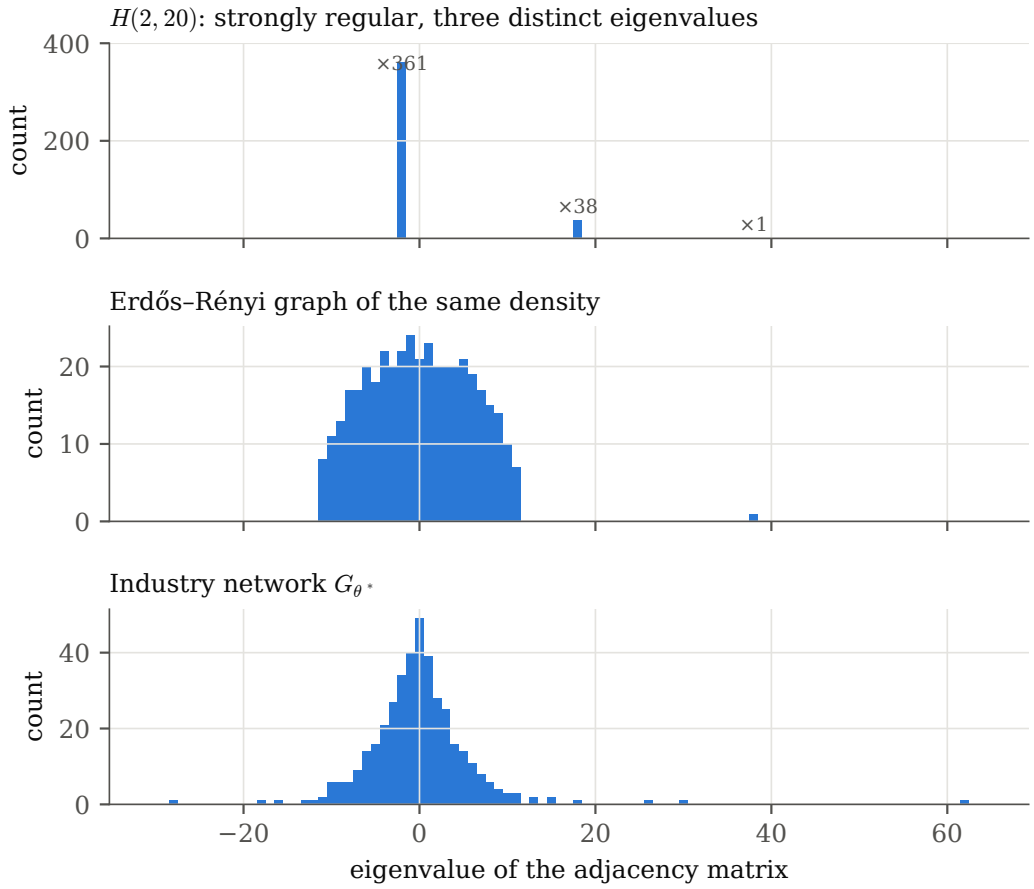


Figure 5: Adjacency spectra of $H(2, 20)$ (three distinct eigenvalues, with multiplicities), of one Erdős–Rényi graph of the same density as G_{θ^*}, and of G_{θ^*}.

Table 2: The three graphs at $\theta^* = 0.0047$. The $H(2, 20)$ column is exact and G_{θ^*} is a computation on the 2017 tables. In the Erdős–Rényi column: (a) expected value, by construction; (b) one sample; (c) the expectation $(N - 2)p^2$ from Proposition 2.10; (d) mean over 200 samples.

	$H(2, 20)$	Erdős–Rényi	G_{θ^*}
Mean degree $\bar{d}$	38	38^a	38.0
Standard deviation of degree	0	5.3^b	37.6
$\bar{\lambda}$	18	3.61^c	17.17
$\bar{\mu}$	2	3.61^c	6.00
$\bar{\lambda} - \bar{\mu}$	16	0.00^d	11.17
δ	0	0.425^d	0.649
Distinct eigenvalues of A	3	400^b	399
Largest eigenvalue of A	38	38.4^b	62.2
Clique number	20	5^b	14

Stability over time. The same analysis on the 2012 tables, at the same level of detail, gives $\theta^* = 0.0048$, a gap $\bar{\lambda} - \bar{\mu} = 10.73$ (degree-preserving upper end 9.46) and $\delta = 0.648$ (random-graph upper end 0.432, degree-preserving lower end 0.663), the same pattern as in 2017. At the coarser level of 71 industries, available for every year from 2012 to 2023, we repeat the analysis at the threshold that matches the density of $H(2, 20)$ (Figure 6). In all 12 years the gap exceeds the degree-preserving range and δ lies between the two random ranges. Across the years $\rho(A)$ stays between 0.471 (2020) and 0.508 (2013), and the average supply-chain length between 1.89 and 2.02 steps.

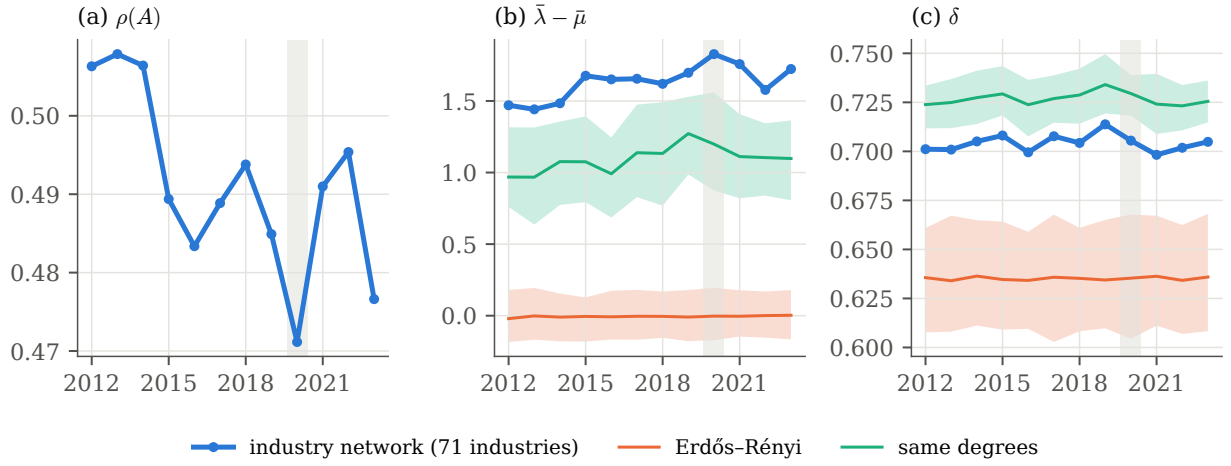


Figure 6: Year by year, 2012–2023, for the 71-industry tables: (a) the spectral radius $\rho(A)$; (b) the gap $\bar{\lambda} - \bar{\mu}$ and (c) δ, at the threshold matching the density of $H(2, 20)$, with the central 95% ranges of the two random benchmarks. The year 2020 is shaded.

Summary. On these tables, supply chains are short: half of all requirements are direct purchases, and the average chain is about two steps long, with longer chains decaying at the rate $\rho(A)$. As a graph, the network is dominated by a few service industries that supply most other industries. Even so, whether two industries trade directly predicts how many trading partners they share more strongly than the unequal degrees alone would produce. What separates the network from strong regularity is variation: the number of shared partners varies greatly from one pair to the next, which accounts for most of δ. These are statements about the US input–output tables over 2012–2023, not general facts about supply chains.

2.7 Limitations

An industry aggregates many firms, so an edge between two industries does not mean that any particular firms trade; firm-level supplier networks are not publicly available. The tables record total requirements, including inputs that were imported, so A describes the technology of US production rather than purely domestic supply chains. Thresholding discards both the size of each requirement and its direction; we report a sweep of thresholds for the first reason, and the directed matrix A itself is used in Section 2.2. Restricting to the 400 largest industries treats the two smallest as outside suppliers; no industry buys more than 3.6 cents of inputs from them per dollar of its output. Finally, the benchmarks are deliberately extreme: a network that sits between them

is neither perfectly uniform nor random, which is informative but is not a model of how supply chains form.

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